PITCHER-HITTER MATCHUP

Hitter: Aaron Judge
RHH, NYY

	2024 Season	MLB Rank
BA	0.307	76th
HR	34	1st
RBI	85	1st
OPS	1.112	1st

Pitcher: Tanner Houck
RHP, BOS

	2024 Season	MLB Rank
W-L	8-6	17th
ERA	2.54	14th
K	112	21st
WHIP	1.03	13th

NOTES

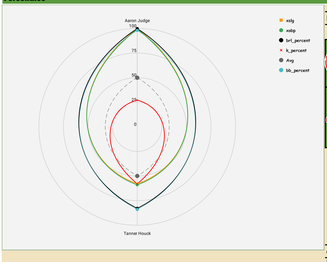
Judge: *Judge is the preeminent power hitter in the game and hits far average as well. He can make solid contact, more consistently than anyone.*

Houck: *Houck is near the top in ERA for the AL, relying on his prevention of hits and walks (3.03 WHIP). 112 Strikeouts are robust, but not enough to call him a deadly strikeout pitcher necessarily.*

GLOSSARY

OPS WHIP ERA

Percentities



NOTES:

The gray dot - denoting 50th percentile- represents MLB avg.

Judge: *Aaron Judge is the top hitter in the MLB in barrel rate* **(100th Perc. Brl%)**, *and ability to hit for extra bases* **(100th Perc. xSLG)**. *He's near the top in getting on base* **(99th Perc. xOBP)**, *driven by his penchance for drawing walks when pitchers fear his power* **(98th Perc. BB%)**, *coupled with his restraint from balls* **(94th Perc. Chase%)**. *His weakness lies in the* **25th percentile strikeout** *ranking, telling us that he must have certain types of pitches in the strikezone that he whiffs on (due to the fact that he rarely chases outside).*

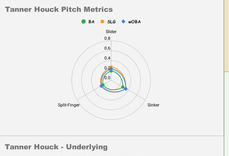
Houck: *Houck's profile appears less impressive off the bat - but his* **58th percentile** *rank in* **strikeouts (+8 percentile points over avg)** *among pitchers is a way to take advantage of Judge* **(-25 pp below avg)**. *He's above average in xOBP allowed and xSLG allowed, but not elite. His elite assets are his deftness in avoiding* **Barrels (hard and high drives, 83rd Perc. Brl%)**, *and Walks, leading to the impressive WHIP number in the section above. He does give up* **hard hits (16th perc. Hard-Hit%)**, *so we can assume that he relies on some combination of location and movement to induce ground balls avoiding barrels, and to avoid walking hitters.*

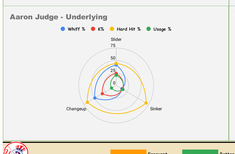
GLOSSARY

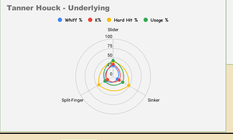
Barrel xSLG% xOBP% K%- strikeout % BB% Chase% Hard-Hit%

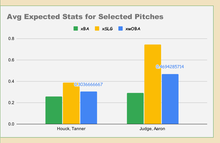
PITCH BREAKDOWN

Select: Pitch All









Frequent | Better Than Average | Worse Than Average

Judge vs MLB Averages

Pitch	Run Value	Pitches	Usage %	BA	xBA	SLG	xSLG	wOBA	xwOBA	Whiff %	K%	Hard Hit %
Slider	1	355	20.8	0.231	0.242	0.387	0.474	0.309	0.356	43.3	24.1	44.2
Sinker	12	271	15.9	0.407	0.353	0.797	0.769	0.551	0.505	13.3	14.3	69.4
Changeup	3	180	10.5	0.172	0.313	0.621	0.625	0.401	0.504	49.5	32.4	66.7

Frequent | Better Than Average | Worse Than Average

Houck vs MLB Averages

Pitch	Run Value	Pitches	Usage %	BA	xBA	SLG	xSLG	wOBA	xwOBA	Whiff %	K%	Hard Hit %	Vis. Mov.	Vert Mov.	Vis Rel	Vert Rel	RPY
Slider	-12	758	42.8	0.191	0.23	0.273	0.402	0.234	0.291	29.4	33	39.7	-17	-1.3	2002	83.1	2.7
Sinker	-2	536	29.9	0.275	0.303	0.344	0.452	0.34	0.362	12.9	18.5	48.1	16.4	0.3	2188	93.4	5.8
Split-Finger	6	456	25.5	0.188	0.237	0.234	0.306	0.228	0.268	23.5	17	43.8	13.6	-3	1783	88.3	3.6

Notes:

Here I've set up a Pitch breakdown, selecting the pitched in Houck's arsenal (Slider, Sinker, Split-Finger) changeup for Judge since he hasn't faced enough Split-Fingers *| expected stats (ex: xBA) : only based on exit velo and launch angle*

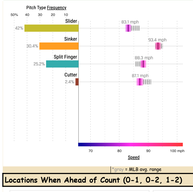
Judge: *The swift majority of Judge's production comes off Fastballs* **(4-Seam Fastball** – *not selected – has a whopping* **Run Value of 35**), *and that's reflected in his uptick in* **batting average (.353) and SLG (.769) on Sinkers**, *sporting a* **70% hard hit rate** *on them. He outperforms league average in SLG, wOBA in all 3 pitches. His one weak point appears to be making contact with* **Sliders** *in the zone,* **(.21 BA, 41% Whiff – below avg)** *and a 20pp decrease in hard hits.*

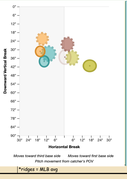
Houck: *Houck's expected stats rank weaker his actuals, meaning he's prone to regress. Houck's most reliable pitch is the* **Slider (42% usage, Run Value 12)**, *which yields an impressive sub .200 BA. Houck doesn't rely on generating Whiffs as we see in the percentile analysis, meaning his reliable slider generates outs by effective location that freezes hitters or gets groundballs. He can take advantage of Judge's weakness though, given that he doesn't become too predictable, by locating his Frisbee* **Slider (17 in movement +140% increase from league avg)** *to freeze him.* **(Houck 33% K on Sliders – above avg)**. *His Hard Hit rate on his slider doesn't look good, which means Judge can crush it if it's left in the heart of the plate* **(83 MPH – below avg)**. *He'll need to mix in his* ***Split-Finger*** *which is also a positive pitch for him* **(Run Value 6)**, *given that his* **Sinker** *is a negative asset* **(Run Value -2)**, *especially against Judge*

Glossary

Vis. Slider xBA xOBA xwOBA

Pitcher Movement and Location





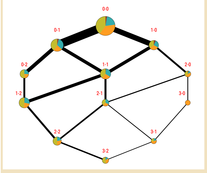
He's thrown an above average % of his pitches on the edge of the zone at 44.5%. League average is 39%.

Average MLB avg. range

*Hrdges = MLB avg

Locations When Ahead of Count (0-1, 0-2, 1-2)

Pitch Mix by Count

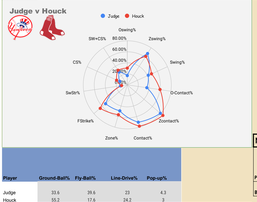


Houck Pitch Movement / Location Notes:

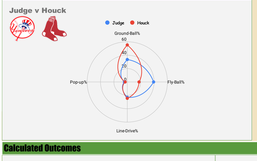
- *Significant Movement on Slider away from Righty Batter -4 pp over avg in Edge strikes.*
- *Above avg movement on Splitter as well, into the Righty Batter*
- *Location moves down when ahead in the count and when generated misses - groundball pitcher signs*
- *Pitch Mix shows leon on Slider when ahead, Sinker when behind*

Plate Discipline / Batted Ball Profiles

Player	Oswing%	Zswing%	Swing%	O-Contact%	Zcontact%	Contact%	Zone%	FStrike%	SwStr%	CS%	SW+CS%
Judge	17.98%	67.48%	41.99%	56.30%	79.20%	70.40%	49.30%	52.30%	12.40%	15.88%	27.40%
Houck	30.30%	61.80%	48.00%	70.90%	83.10%	79.20%	56.30%	65.60%	10.50%	21.60%	31.50%



Player	Ground-Ball%	Fly-Ball%	Line-Drive%	Pop-up%
Judge	33.6	39.6	23	4.3
Houck	55.2	17.6	24.2	3

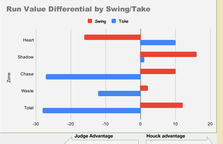


Notes:

Plate Discipline: *A glaring difference in their plate discipline profiles come in* **First Pitch Strikes (Houck: 65%, Judge: 52%)**. *Houck is likely to test Judge in the strikezone more than most pitchers dare to* **(Houck: 56% Zone)**. *If he can steal strikes on the edges with his looping slider early, he might be able to induce a rare chase from Judge later. On the other hand, Houck gives up a decent amount of contact on all swings* **(78%)**, *which could favor Judge. Houck's pounding of the strikezone leads to* **10% more strikes looking** *than swinging.*

Batted Balls: *As we suspected before, Houck is a groundball pitcher due to his slider that breaks below barrels, and Judge is obviously a fly ball hitter who might wait for the Split Finger or Sinker to pull into the air.*

Calculated Outcomes



Better than Original Houck Outcome Probability | Worse than initial Houck Outcome Probability

Houck v Judge Outcome Probabilities*

Pitches	Whiff Prob	K Prob	BB Prob	Hard Hit Prob	xBA	xSLG	Run Value (+ for Houck)
Slider	33.25%	27.89%	6.80%	41.14%	0.22	0.43	11
Sinker	17.87%	21.62%	5.90%	56.30%	0.32	0.61	-14
Split Change	39.28%	33.33%	4.25%	33.33%	0.28	0.56	8
Overall	**27.25%**	**25.53%**	**6.26%**	**47.19%**	**0.25**	**0.51**	**5**
Margin of Error (95% confidence)	3%	6%	2%	6%	7%	5%	

*Probabilities calculated by combining the pitcher and the hitter's stats, placing more weight on more frequent pitches. Also leverages judge's performance against pitch movement specific to Houck

92% of Houck Sliders move between 8 and 20 inches glove-side, 80% of Houck Sinkers and Splitters move between 12 and 18 inches arm-side

Final Analysis

Houck is Unlikely to walk Judge **(6% chance)**, even though Judge usually garners lots of balls. Houck's ability to find the edge of the strikezone with his high-movement slider is his biggest asset, as Judge doesn't make contact as well on those types of pitches **(.22 expected BA v Houck, 27% Whiff chance)**. Houck has a tendency to give up hard hit balls which could come into play against Judge **(48% chance)**, as he must keep his location low to avoid that, as he does usually with his effective Slider. The vigilance must be exercised especially with the Sinker **(56% Hard Hit Probability)**. Notably, in the Run Value differential chart by pitch, Houck accumulates strikes with Sliders in the Heart on the plate **(+10)**, but gives up a lot of his leverage when Sinkers are put in the zone **(-11)**. The pitch matchup by Run Value is dead even due to Houck's large on-paper advantage on the **Slider (+11)** balanced out by Judge's edge on the **Sinker (-14)**

7/19/2024